STPM 2021 PHYSICS 960/3

Questions and Answers

Values of constants: $h = 6.63 \times 10^{-34}$ J s, $c = 3.00 \times 10^{8}$ m s^{-1}, $e = 1.60 \times 10^{-19}$ C, $m_e = 9.11 \times 10^{-31}$ kg, 1 u $= 1.66 \times 10^{-27}$ kg.

Section A [15 marks]

1. Which statement is true about simple harmonic motion?

A The momentum is constant.
B The kinetic energy is constant.
C The restoring force is constant.
D The angular frequency is constant.

Answer: D

In SHM the angular frequency ω (and so the period) is constant, independent of amplitude.

Explanation: The momentum, kinetic energy and restoring force all vary with the position of the particle.

2. An object of mass 200.0 g is attached to a spring with its spring constant of 500.0 N m^{-1} on a horizontal smooth surface. The spring is stretched to 0.045 m and released from its resting state. What is the total mechanical energy at the equilibrium position?

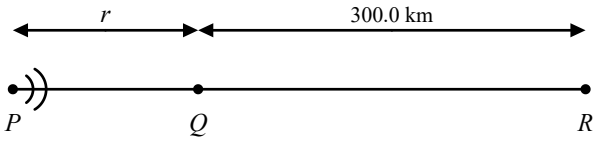
A 0 B 0.010 J C 0.51 J D 11 J

Answer: C

$$E = \frac{1}{2}kA^2 = \frac{1}{2}(500.0)(0.045)^2 = 0.51 \text{ J}$$

Explanation: The total mechanical energy is constant everywhere in the motion; at equilibrium it is all kinetic.

3. Q is located in between an earthquake wave source, P, and R. The distance between Q and P is r and the distance between Q and R is 300.0 km as shown in the diagram below.



The intensities of the waves at Q and R are 2.0×10^6 W m^{-2} and 5.0×10^4 W m^{-2} respectively. If P is assumed to be the point source, what is the value of r?

A 7.7 km B 47 km C 56 km D 120 km

Answer: C

$$I \propto \frac{1}{d^2} \Rightarrow \frac{(r+300)^2}{r^2} = \frac{2.0 \times 10^6}{5.0 \times 10^4} = 40$$

$$\frac{r+300}{r} = \sqrt{40} = 6.32 \Rightarrow r = 56 \text{ km}$$

Explanation: R is $(r + 300)$ km from the source. For a point source the intensity falls as the inverse square of distance.

4. Which statement is **not** true about mechanical and electromagnetic waves?

A The mechanical and electromagnetic waves undergo Doppler effect.
B The mechanical waves and all electromagnetic waves could be plane polarised.
C The mechanical waves require a medium for propagation but the electromagnetic waves do not require any medium for propagation.
D All electromagnetic waves propagate with the same speed in vacuum but the mechanical waves propagate with different speeds in different medium.

Answer: B

Longitudinal mechanical waves (e.g. sound) cannot be plane polarised; only transverse waves can. Statement B is not true.

Explanation: All electromagnetic waves are transverse and can be polarised, but not all mechanical waves.

5. An ambulance approaches a car which is moving in an opposite direction. The speed of the ambulance is 40.0 m s^{-1}. The siren of the ambulance emits sound of frequency 500 Hz. What is the frequency of the sound heard by the driver of the car if the car moves at 18.0 m s^{-1} towards the ambulance?
[Assume that the speed of sound is 345 m s^{-1}.]

A 425 Hz B 471 Hz C 536 Hz D 595 Hz

Answer: D

$$f' = f\frac{v + v_o}{v - v_s} = 500\frac{345 + 18.0}{345 - 40.0} = 595 \text{ Hz}$$

Explanation: Source and observer move towards each other, so both effects raise the frequency.

6. A string is fixed at both ends. Then, the string is plucked in the middle. The length, mass and tension of the string are 2.0 m, 8.0 g and 1.2 kN respectively. What is the fundamental frequency of the string?

A 4.3 Hz B 68 Hz C 140 Hz D 270 Hz

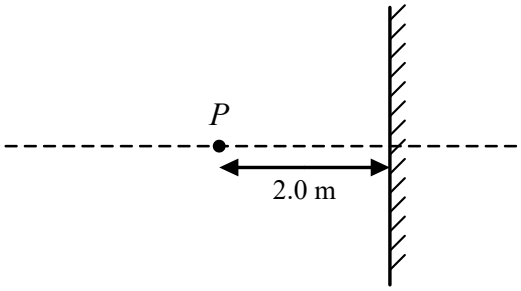
Answer: C

$$\mu = \frac{8.0 \times 10^{-3}}{2.0} = 4.0 \times 10^{-3} \text{ kg m}^{-1}; v = \sqrt{\left(\frac{1200}{4.0 \times 10^{-3}}\right)} = 548 \text{ m s}^{-1}$$

$$f = \frac{v}{2L} = \frac{548}{4.0} = 137 \text{ Hz} \approx 140 \text{ Hz}$$

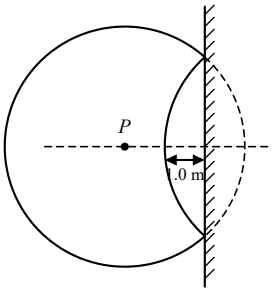
Explanation: Plucking in the middle excites the fundamental, with $L = \lambda/2$. Tension 1.2 kN = 1200 N.

7. A source of light, *P*, is placed 2.0 m in front of a plane mirror as shown in the diagram below.

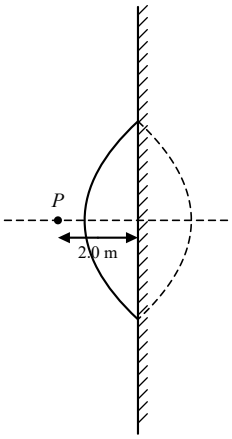


Which diagram shows the correct wave front of the light after 1.0×10^{-8} s the light flashed?

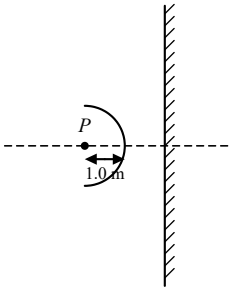
A



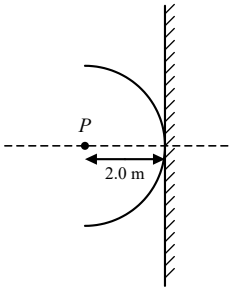
B



C



D



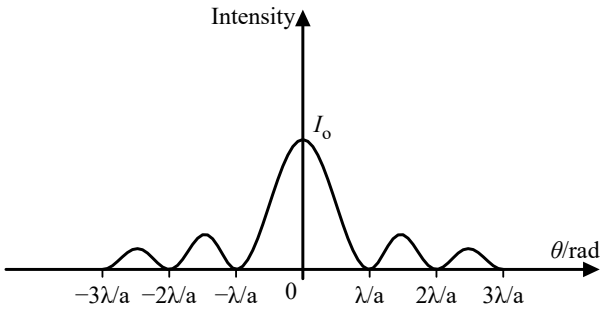
Answer: A

In 1.0×10^{-8} s light travels $(3.0 \times 10^{8})(1.0 \times 10^{-8}) = 3.0$ m

The wavefront is a sphere of radius 3.0 m about *P*; the part that reached the mirror (2.0 m away) is reflected and has travelled back 1.0 m, as in A.

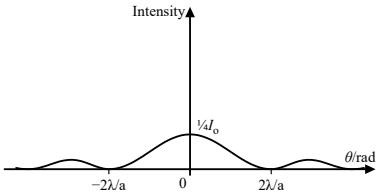
Explanation: The reflected wavefront appears to come from the image of *P*, 2.0 m behind the mirror.

8. The variation of light intensity with the angle of diffraction, θ, through a single slit of width, a, is shown in the diagram below.

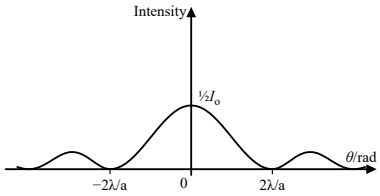


Which diagram shows the correct intensity distribution when a is halved?

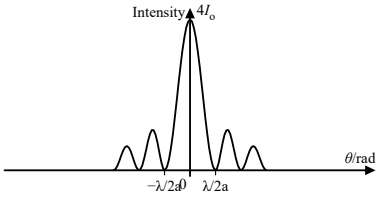
A



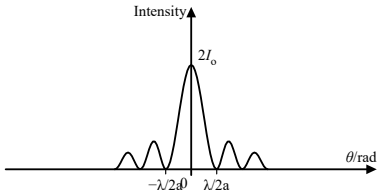
B



C



D



Answer: A

Halving the slit width doubles the angular width of the central maximum (first minima at $\pm 2\lambda/a$). The amplitude at the centre is proportional to a, so the central intensity falls to $\frac{1}{4}I_o$ (A).

Explanation: Less light passes through the narrower slit and it is spread over a wider angle.

9. Light from a He-Ne laser of wavelength 632 nm strikes a diffraction grating that has 600 lines per milimeter. What is the maximum number of the bright fringes that could be observed after the strikes?

A 3 B 4 C 5 D 6

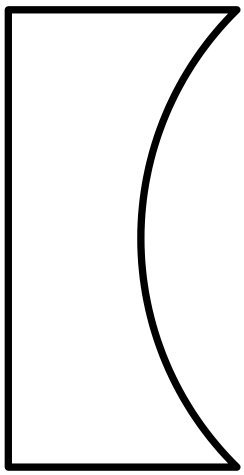
Answer: C

$$d = \frac{1}{600 \times 10^3} = 1.67 \times 10^{-6}\ \text{m};\ \frac{d}{\lambda} = \frac{1.67 \times 10^{-6}}{632 \times 10^{-9}} = 2.6 \Longrightarrow n_{\max} = 2$$

Bright fringes: $n = 0, \pm 1, \pm 2 \Longrightarrow$ 5 fringes.

Explanation: Count the central maximum and the orders on both sides.

10. A thin lens of power −2.1 D has a flat and a curved surfaces as shown in the diagram below.



If the refractive index of the lens is 1.63, what is the radius of curvature of the lens?

A −40.0 cm B −30.0 cm C 30.0 cm D 40.0 cm

Answer: B

$$P = (n-1)\left(\frac{1}{r_1} + \frac{1}{r_2}\right) \text{ with } r_1 = \infty: -2.1 = 0.63\frac{1}{r}$$

$$r = -0.30 \text{ m} = -30.0 \text{ cm}$$

Explanation: The negative power shows a diverging lens; with the real-is-positive convention, a concave surface has a negative radius.

11. An electron and a photon have the same momentum. If the speed of light is c, the mass of the electron is m and the kinetic energy is E, what is the energy of the photon?

A $\frac{\sqrt{(mEc^2)}}{2}$ B $\sqrt{(mEc^2)}$ C $2\sqrt{(mEc^2)}$ D $\sqrt{(2mEc^2)}$

Answer: D

Electron: $p = \sqrt{(2mE)}$; photon: $E_{\text{photon}} = pc = \sqrt{(2mEc^2)}$

Explanation: A photon's energy and momentum are related by $E = pc$.

12. The first postulate of Niels Bohr on atomic model states that electrons of an atom surround the nucleus in certain orbits. What can be deduced from the postulate?

A The energy of the electrons in the orbit is always constant.
B The accelerating electrons generate electromagnetic waves.
C The electrons could be in any orbits and have any amount of energy.
D The electrons which surround the nucleus will eventually fall into the nucleus.

Answer: A

In an allowed (stationary) orbit the electron does not radiate, so its energy remains constant.

Explanation: Bohr's postulate contradicts classical physics, which predicts that an accelerating electron radiates and spirals into the nucleus.

13. Which statement is true about X-ray?

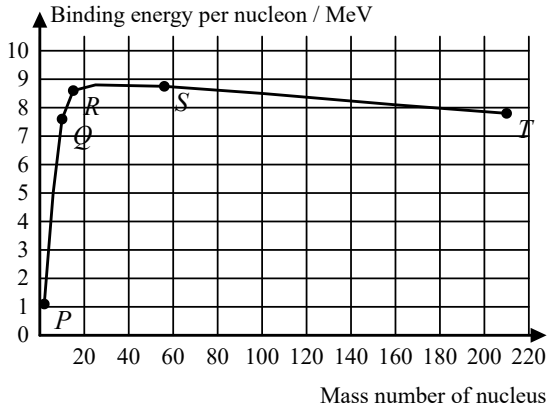
A X-ray is produced when electrons are ejected from hot metal surface.
B X-ray is an electromagnetic wave of wavelength approximately 10^{-7} m.
C Higher potential difference between the cathode and anode produces X-ray with lower penetrating power.
D Higher potential difference between the cathode and anode produces shorter wavelength in X-ray spectrum.

Answer: D

$\lambda_{min} = \frac{hc}{eV}$: a higher voltage gives a shorter minimum wavelength

Explanation: X-rays are produced when fast electrons hit a target (not thermionic emission); typical wavelengths are about 10^{-10} m; a higher voltage gives more penetrating (harder) X-rays.

14. The variation of nuclear binding energy per nucleon with the mass number of nucleus for nuclei, *P*, *Q*, *R*, *S* and *T*, is shown in the graph below.



Which is **not** true about the above nuclei?

A *S* is the most stable nucleus.
B *R* could decay to *Q* spontaneously.
C *T* could split into two lighter nuclei and energy is released.
D Two *P* could fuse to form a heavier nucleus and energy is released.

Answer: B

Q has a lower binding energy per nucleon than *R*. A change from *R* to *Q* would need energy, so it cannot happen spontaneously. Statement B is not true.

Explanation: Nuclei tend to change towards a higher binding energy per nucleon: fusion of light nuclei (*P*) and fission of heavy nuclei (*T*) release energy; *S* near the peak is most stable.

15. A sample contains 1.0×10^{22} nuclei of a radioactive isotope. If a total of 9.7×10^{21} nuclei in the sample decayed in 60 s, what is the half-life of the isotope?

A 5.2 s B 12 s C 31 s D 58 s

Answer: B

$$\text{Remaining} = 1.0 \times 10^{22} - 9.7 \times 10^{21} = 3.0 \times 10^{20};\ \frac{N}{N_0} = 0.030$$

$$(\frac{1}{2})^{t/T} = 0.030 \Longrightarrow \frac{t}{T} = 5.06 \Longrightarrow T = \frac{60}{5.06} = 12 \text{ s}$$

Explanation: Use the number remaining, not the number decayed.

Section B [15 marks]

16. A bomb blasts 12 m perpendicularly from the right eardrum of a soldier. The eardrum of the soldier has an area of 25 mm^2. The sound energy received by the soldier in every second is 1.0×10^{-9} W.

[Assume that the threshold of hearing is 10^{-12} W m^{-2}.]

(a) Determine the level of the sound intensity at the soldier's eardrum. [3]

Answer

$I = \frac{P}{A} = \frac{1.0 \times 10^{-9}}{25 \times 10^{-6}} = 4.0 \times 10^{-5}$ W m^{-2} — 1

$\beta = 10 \log \frac{I}{I_0} = 10 \log \frac{4.0 \times 10^{-5}}{10^{-12}}$ — 1

$\beta = 76.0$ dB — 1

3

Explanation: 25 $mm^2 = 25 \times 10^{-6}$ m^2.

(b) Determine the change in the level of the sound intensity if the soldier is 17 m from the blast. [3]

Answer

$I \propto \frac{1}{r^2} \Longrightarrow \Delta\beta = 10 \log (\frac{12}{17})^2$ — 1

$\Delta\beta = -3.0$ dB — 1

The level decreases by 3.0 dB (to 73.0 dB) — 1

3

Explanation: Moving from 12 m to 17 m roughly halves the intensity, which is a drop of 3 dB.

(c) If the level of the sound intensity is increased by three decibels, what will happen to the sound intensity? [1]

Answer

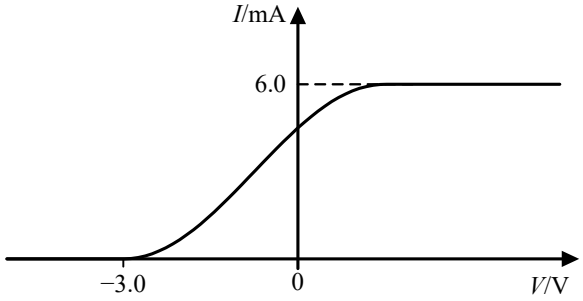
The sound intensity doubles ($\times 10^{0.3} \approx 2$) 1

1

Total for question **7**

Explanation: +3 dB corresponds to an intensity ratio of $10^{0.3} \approx 2$.

17. In an experiment to study the photoelectric effect, the variation of photoelectric current, I, with potential difference, V, across the cathode and anode is shown in the graph below.



Monochromatic light of wavelength 250 nm is irradiated on the surface of the cathode.

(a) Determine the maximum speed of the photoelectrons. [2]

Answer

From the graph $V_s = 3.0$ V: $\frac{1}{2}mv^2 = eV_s$ 1

$v = \sqrt{\left(\frac{2(1.60 \times 10^{-19})(3.0)}{9.11 \times 10^{-31}}\right)} = 1.03 \times 10^6$ m s^{-1} 1

2

Explanation: The stopping potential is where the current first falls to zero on the negative side.

(b) Calculate the threshold frequency of the incident radiation to produce photoelectrons. [3]

Answer

$W = \frac{hc}{\lambda} - eV_s = \frac{(6.63 \times 10^{-34})(3.00 \times 10^8)}{250 \times 10^{-9}} - (1.60 \times 10^{-19})(3.0)$ 1

$W = 7.96 \times 10^{-19} - 4.80 \times 10^{-19} = 3.16 \times 10^{-19}$ J 1

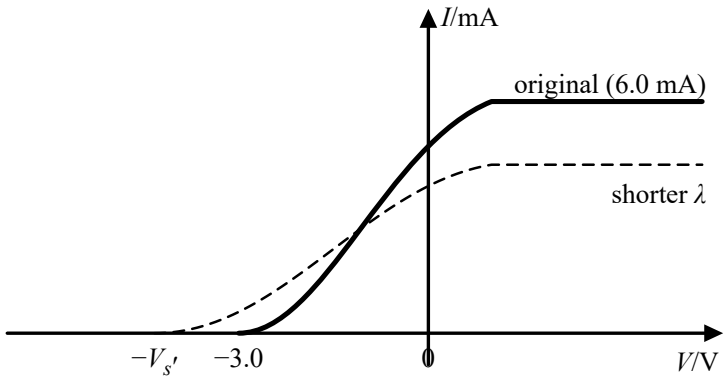
$f_0 = \frac{W}{h} = \frac{3.16 \times 10^{-19}}{6.63 \times 10^{-34}} = 4.76 \times 10^{14}$ Hz 1

3

Explanation: The threshold frequency is the frequency whose photon energy equals the work function.

(c) If the experiment is repeated using the monochromatic light of shorter wavelength but of the same intensity, sketch a curve to show the variation of I with V on the axes of the graph above. [3]

Answer



Larger stopping potential (curve meets the axis further left than −3.0 V)	1
Lower saturation current	1
Same intensity with higher-energy photons means fewer photons per second, so fewer photoelectrons, but each has more kinetic energy	1
	3
Total for question	**8**

Explanation: Intensity = (number of photons per second per unit area) × (energy per photon).

Section C [30 marks]

18. (a) State **three** differences between progressive and standing waves. [3]

Answer

• Progressive waves transfer energy; standing waves do not transfer energy	1
• In a progressive wave all particles have the same amplitude; in a standing wave the amplitude varies from zero (nodes) to maximum (antinodes)	1
• In a progressive wave neighbouring particles have different phases; in a standing wave all particles between two adjacent nodes are in phase	1
	3

Explanation: Other acceptable differences: a standing wave has fixed nodes and antinodes; the wave profile of a progressive wave moves.

(b) A wave propagates in a string with an amplitude of 0.35 m, wavelength of 0.667 m and frequency of 5.00 Hz.

[Given that $\sin A + \sin B = 2 \sin (A + B)/2 \cos (A - B)/2$ and $\cos A + \cos B = 2 \cos (A + B)/2 \cos (A - B)/2$.]

(i) Determine the expression for the wave. [4]

Answer

$\omega = 2\pi f = 10\pi$ rad s^{-1}	1
$k = \frac{2\pi}{\lambda} = \frac{2\pi}{0.667} = 3\pi$ m^{-1}	1
$y = A \sin (\omega t - kx)$ for a wave travelling in the $+x$ direction	1
$y = 0.35 \sin (10\pi t - 3\pi x)$ m	1
	4

Explanation: Taking $y = 0$ at $x = 0$ and $t = 0$; the minus sign gives propagation in the positive x direction.

(ii) Determine the velocity of the wave. [2]

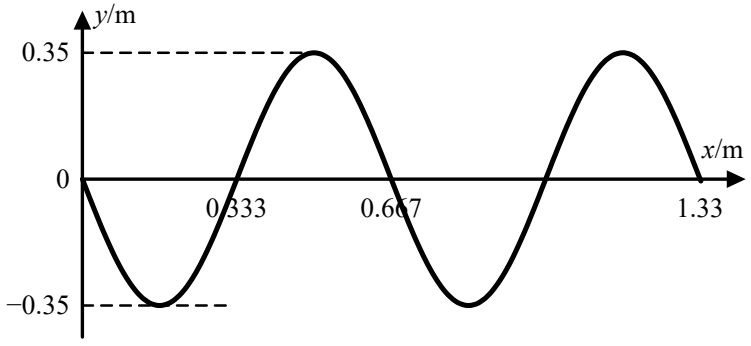
Answer

$v = f\lambda = (5.00)(0.667)$	1
$v = 3.34$ m s^{-1} (in the $+x$ direction)	1
	2

Explanation: Equivalently $v = \omega/k = 10\pi/3\pi = 3.33$ m s^{-1}.

(iii) Sketch and label a graph for the displacement against distance of the wave at $t = 0$. [3]

Answer



Sine curve of amplitude 0.35 m	1
Wavelength 0.667 m marked	1
Correct starting phase: $y = -0.35 \sin 3\pi x$ (starts at 0, going negative)	1
	3

Explanation: At $t = 0$ the expression becomes $y = 0.35 \sin (-3\pi x) = -0.35 \sin (3\pi x)$.

(iv) Then, the wave superposed with an identical wave which propagates in the opposite direction. Derive the expression for the resultant wave. [3]

Answer

$y = 0.35 \sin(10\pi t - 3\pi x) + 0.35 \sin(10\pi t + 3\pi x)$ 1

$= 2(0.35) \sin \frac{(10\pi t - 3\pi x) + (10\pi t + 3\pi x)}{2} \cos \frac{(10\pi t - 3\pi x) - (10\pi t + 3\pi x)}{2}$ 1

$y = 0.70 \cos(3\pi x) \sin(10\pi t)$ m — a standing wave 1

3

Total for question **15**

Explanation: $\cos(-3\pi x) = \cos(3\pi x)$. The amplitude $0.70 \cos(3\pi x)$ depends on position, the mark of a standing wave.

19. (a) A sharp image of a distant object is formed on a sheet of paper placed near to a clear glass ball of diameter 15.0 cm and refractive index 1.50. Determine the distance of the image from the side of the ball that faces the object. [6]

Answer

$r = 7.50$ cm. First surface, object at infinity: $\frac{1.50}{v_1} = \frac{1.50 - 1}{7.50}$ 1

$v_1 = 22.5$ cm (beyond the first surface) 1

This image is $22.5 - 15.0 = 7.5$ cm beyond the second surface: virtual object, $u_2 = -7.50$ cm, $r_2 = -7.50$ cm 1

$\frac{1.50}{-7.50} + \frac{1}{v_2} = \frac{1 - 1.50}{-7.50}$ 1

$v_2 = 3.75$ cm beyond the far side of the ball 1

Distance from the side facing the object $= 15.0 + 3.75 = 18.75$ cm 1

6

Explanation: Treat the ball as two refracting surfaces. The image from the first surface acts as a (virtual) object for the second.

(b) If the clear glass ball in (a) is reshaped into a biconvex thin lens with radii of curvatures r_1 and r_2, derive a formula to show the relationship between the distance of the object, u, the distance of the image, v, and the focal length, f. [4]

Answer

First surface: $\frac{1}{u} + \frac{n}{v_1} = \frac{n-1}{r_1}$ 1

Second surface (thin lens, $u_2 = -v_1$): $-\frac{n}{v_1} + \frac{1}{v} = \frac{n-1}{r_2}$ 1

Adding: $\frac{1}{u} + \frac{1}{v} = (n-1)(\frac{1}{r_1} + \frac{1}{r_2})$ 1

For an object at infinity $v = f$, so $\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$ 1

4

Explanation: For a biconvex lens with this sign convention both radii are taken as positive, giving the lensmaker's equation.

(c)

(i) A biconvex thin lens has equal radius of curvature 15.0 cm. Determine the distance of the image when an object is placed 60.0 cm in front of the lens. [3]

Answer

$\frac{1}{f} = (1.50-1)(\frac{1}{15.0} + \frac{1}{15.0}) \Rightarrow f = 15.0$ cm 1

$\frac{1}{v} = \frac{1}{15.0} - \frac{1}{60.0}$ 1

$v = 20.0$ cm 1

3

Explanation: The refractive index is the same as the glass ball, 1.50.

(ii) Determine the characteristics of the image formed. [2]

Answer

Real and inverted 1

Diminished ($m = \frac{20.0}{60.0} = \frac{1}{3}$) 1

2

Total for question **15**

Explanation: The object is beyond $2f$ (30 cm), so the image is between f and $2f$.

20. (a) State the law of conservation of nucleon number in a nuclear reaction. [1]

Answer

The total number of nucleons (protons + neutrons) before a nuclear reaction equals the total number after it — 1

1

Explanation: The total charge (proton number) is also conserved.

(b) A $^{226}_{88}\text{Ra}$ nucleus undergoes alpha decay and transforms into a $^{222}_{86}\text{Rn}$ nucleus.

[Mass of $^{226}_{88}\text{Ra}$ is 226.025410 u, mass of $^{222}_{86}\text{Rn}$ is 222.017576 u, mass of helium is 4.002603 u, 1 u ≡ 931.5 MeV.]

(i) Write the equation of the nuclear reaction for the radioactive decay. [1]

Answer

$^{226}_{88}\text{Ra} \rightarrow {}^{222}_{86}\text{Rn} + {}^{4}_{2}\text{He}$ — 1

1

Explanation: Nucleon and proton numbers balance on both sides.

(ii) Determine the energy released in the nuclear reaction. [3]

Answer

$\Delta m = 226.025410 - (222.017576 + 4.002603)$ — 1

$\Delta m = 0.005231$ u — 1

$E = (0.005231)(931.5) = 4.87$ MeV — 1

3

Explanation: Atomic masses can be used because the electron numbers balance.

(iii) Calculate the velocity of the emitted alpha particle. [4]

Answer

Assume all the energy becomes kinetic energy of the α particle (recoil of Rn neglected) — 1

$E = (4.873 \times 10^{6})(1.60 \times 10^{-19}) = 7.80 \times 10^{-13}$ J — 1

$v = \sqrt{\left(\frac{2(7.80 \times 10^{-13})}{(4.002603)(1.66 \times 10^{-27})}\right)}$ — 1

$v = 1.53 \times 10^{7}$ m s^{-1} — 1

4

Explanation: Strictly, momentum conservation gives the α particle about 222/226 of the energy, so its speed is very slightly lower.

(c) A nuclear power plant operates on the principle of sustained chain reaction in its nuclear reactor. To maintain the nuclear reaction, the reactor requires moderator, control rods and fuels.

(i) Explain the chain reaction in the nuclear reactor. [2]

Answer

A U-235 nucleus absorbs a slow neutron and splits, releasing energy and 2–3 neutrons	1
These neutrons cause further fissions; in a reactor the reaction is controlled so that on average exactly one neutron from each fission causes another fission (steady, sustained chain reaction)	1
	2

Explanation: If more than one neutron per fission caused further fission, the reaction would grow uncontrollably.

(ii) State the function of the moderator and control rods. Give an example of a material used for each component. [4]

Answer

Moderator: slows down fast neutrons so they are more likely to cause fission	1
Example: graphite, water or heavy water	1
Control rods: absorb excess neutrons to control the rate of reaction	1
Example: boron or cadmium	1
	4
Total for question	**15**

Explanation: Pushing the control rods further into the core absorbs more neutrons and slows the reaction.